

HDI INSIGHT AI

SYSTEM USER MANUAL & INTERFACE GUIDE

Author: Mohammad Tayyab

Software Version: v1.0.0

Target Audience: Policy Advisors & Researchers

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1. Introduction

Welcome to the **HDI Insight AI** Human Development Analytics Platform. This system is an AI-powered portal designed to predict, analyze, and advise on human development indicator performance. By mapping four essential socioeconomic parameters, the platform provides real-time estimates of a country's Human Development Index (HDI) category along with policy recommendations to improve societal indicators. This document provides step-by-step guidance to operate all integrated user features.

2. Accessing the Application

Once installed and initialized according to the Installation Guide, launch your modern web browser (Chrome, Firefox, or Edge) and navigate to the host port address: **http://localhost:5000/**.

You will be greeted with the main interactive analytics interface. The portal uses clean CSS variables allowing you to instantly toggle between light and dark modes via the toggle in the navigation bar.

3. Interactive Features & Guide

HDI Prediction Simulator

Navigate to the main form to run a new simulation. Modify the input sliders representing the four key indicators: Life Expectancy at birth (years), Expected years of schooling, Mean years of schooling, and Gross National Income (GNI) per capita (PPP \$). Click the **Predict HDI** button to submit. The system will display the predicted HDI category (Low, Medium, High, or Very High) on the confidence gauge, along with the continuous score estimation.

Explainable AI & Recommendations

The model automatically generates local explainability (XAI) diagnostics, plotting the relative weight of each feature on the prediction outcome. Additionally, the intelligent advisor engine provides active development policy advice by comparing your profile variables against standard world medians, highlighting critical focus areas for national investment.

Country Comparisons

Click on the **Country Comparison** tab. Select any real-world country from the dropdown to compare your simulated socioeconomic profile side-by-side with official global datasets. Comparison bar charts will highlight gaps in healthcare, education, and income index factors.

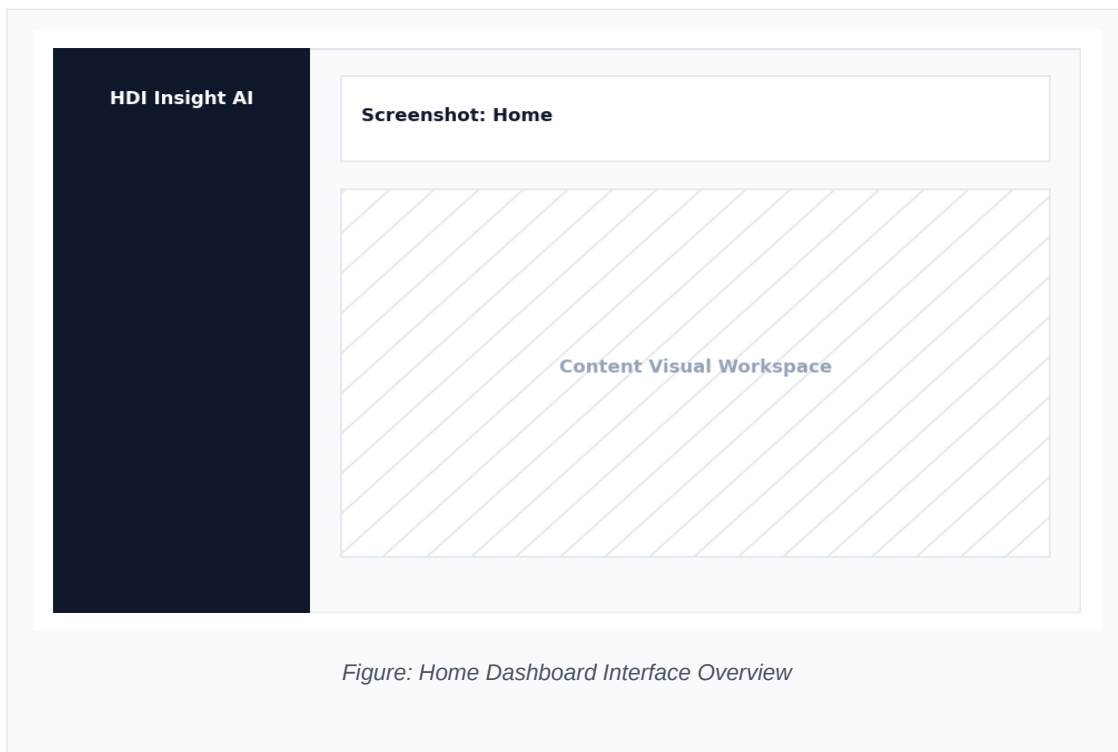
Exporting Reports & History

The system records all successful runs. You can view, search, and delete entries from the prediction log history. Click **Download PDF** on any prediction block to generate a clean, stylized, single-page summary report. Click **Export CSV** at the top of the history list to save the full simulation history database locally.

4. Screen Gallery & Walkthrough

Home Landing Portal

The home page displays the dashboard dashboard, historical logs, and shortcuts to launch a new simulator run.



Prediction Simulator Panel

The simulator page features range input sliders, the primary confidence gauge, and companion regressor outputs.

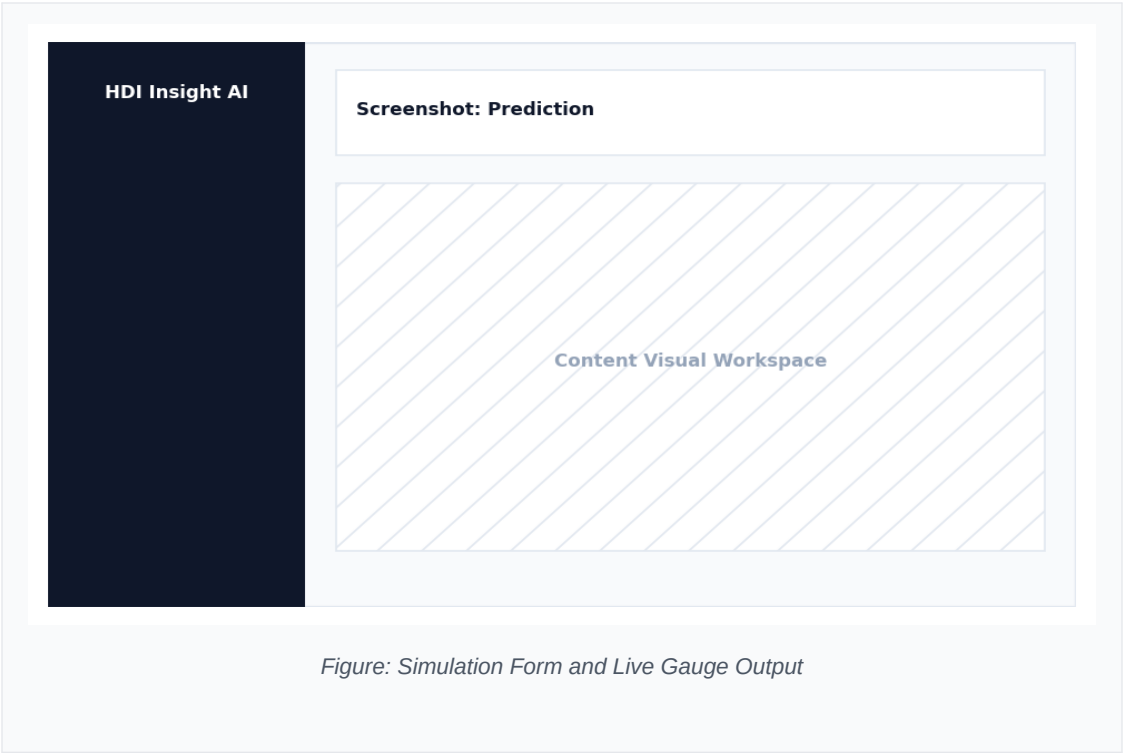


Figure: Simulation Form and Live Gauge Output

Interactive Dashboard

Analytical views displaying summary charts of prediction records and distribution frequencies.

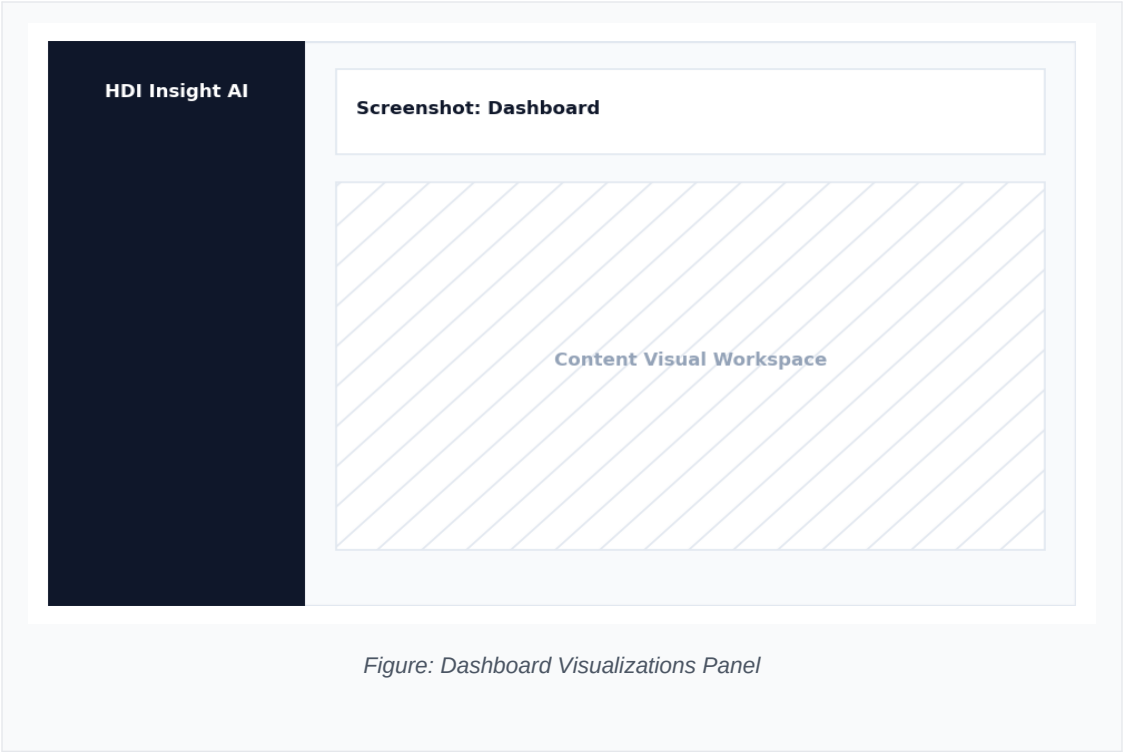
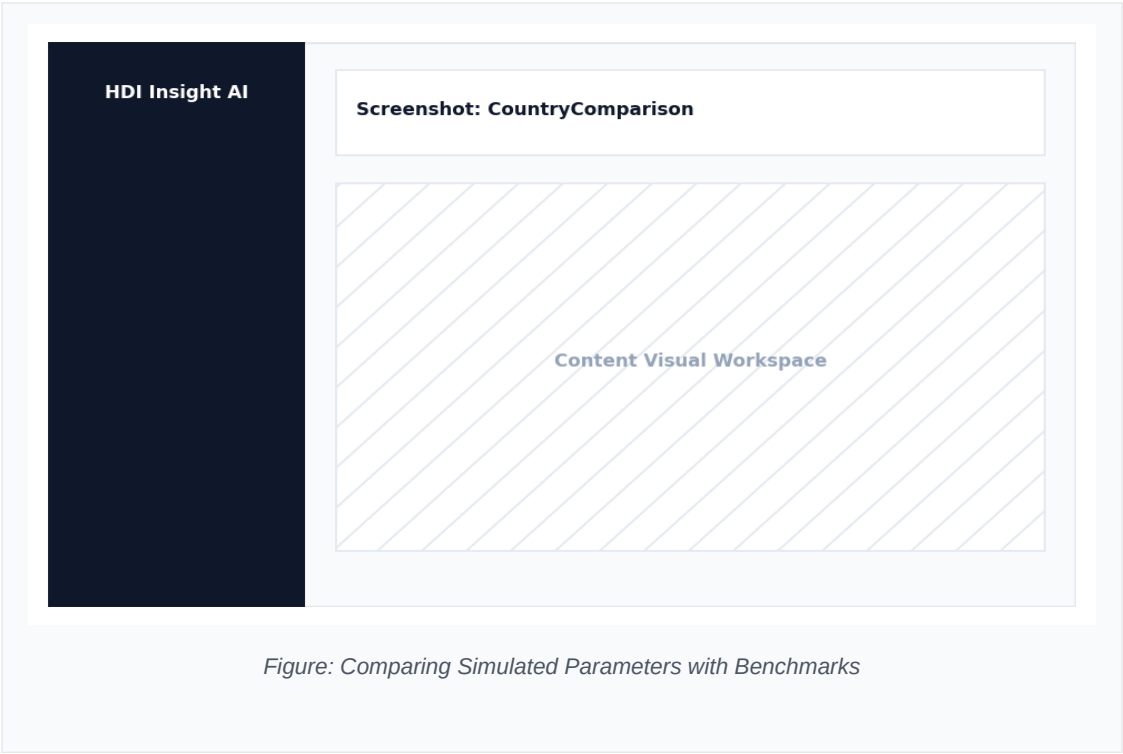


Figure: Dashboard Visualizations Panel

Country Comparison Tool

This panel plots comparative charts matching user-defined profiles against real-world countries.



Model Performance Dashboard

Displays performance metrics tables, model training times, and confusion matrices for all algorithms.

